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CSA1501 - Short Answer Test

1. Define cloud computing and explain any four characteristics.

Cloud Computing is the delivery of computing services, storage, databases, networking, and software over the internet. It allows users to access resources anytime without owning physical infrastructure.

Characteristics:-

1. on-Demand Self-service - users can create or remove resources whenever needed without contacting the provider.
2. Broad network Access - cloud services are available through the internet from laptops, mobiles, and tablets.
3. Resource pooling - A cloud provider shares computing resources among multiple users securely.
4. Rapid Elasticity - Resources automatically increase or decrease based on workload.

Difference from Traditional computing:-

- * Traditional Systems require purchasing and maintaining hardware.
- * cloud computing provides resources online with better flexibility and lower cost.

Ex:- A company hosts its website on AWS instead of buying its own servers.

2. Evolution of cloud computing.

Cloud computing evolved through different stages.

1. Hardware Evolution - large mainframe computers were used by many users.
2. Virtualization - one physical server was divided into multiple virtual machines, improving resource utilization.
3. Internet Evolution - High-speed internet made remote access possible from anywhere.
4. Web Software Evolution - Applications became available through web browsers without installation.
5. Cloud platforms - Companies like AWS, Microsoft Azure, and Google Cloud now provide scalable cloud services on demand.

DevOps view:-

Virtualization and automation made continuous integration (CI) and Continuous deployment (CD) possible, enabling faster software delivery.

3. Role of server virtualization in cloud computing.

Server Virtualization is the process of creating multiple virtual servers on one physical server using a hypervisor.

Benefits:-

- * Better hardware utilization.
- * Lower infrastructure cost.
- * Easy backup and recovery.
- * Faster deployment of applications.

Difference

Virtualization

- * Creates virtual machine

- * Technology

- * Improves hardware usage

Ex- VMware virtual machines, while AWS EC2 provides cloud virtual servers.

Cloud Computing

- * Delivers services over the internet.

- * Service model

- * Provides scalable computing services.

4. Role of Data Centers in Delivering cloud services.

A Data Center is a secure facility that contains servers, storage devices, networking equipment, and cooling systems.

Roles:-

- * Stores customer data securely.

- * Runs cloud applications continuously.

- * Provides backup and disaster recovery.

- * Ensures high availability with minimum downtime.

- * Delivers cloud services to users worldwide.

Devops tools:

Modern data centers use Docker, Kubernetes, and automation tools to deploy applications quickly and reliably.

Ex- AWS data centers host websites, databases, and applications for millions of users.

5 Differentiate IaaS, PaaS, SaaS and XaaS

Service Model	Description	Real-world Example
IaaS	Provider Virtual Servers, Storage, and networking	AWS EC2
PaaS	Provider a platform to build and deploy application	Google App Engine
SaaS	Provider Software over the Internet without installation	Google Docs
XaaS	"Anything as a service" provides any IT service through the cloud	Microsoft Azure

Key Differences.

- * IaaS : You manage applications and operating system.
- * PaaS : You manage only the applications ; the provider manages the platform.
- * SaaS : The provider manages everything.
- * XaaS : A broad model that includes all cloud-based services.